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VLM-Driven Neuro-Symbolic Task and Motion Planning with Ambiguity Resolution

Candidato:
Vincenzo Palomba

Relatori:
Prof. Alberto Finzi

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Outline

1. Introduction
2. Deliberative Layer
3. Executive Layer
4. Reactive Layer
5. Setup and Experimental Results
6. Conclusion

Project Objective

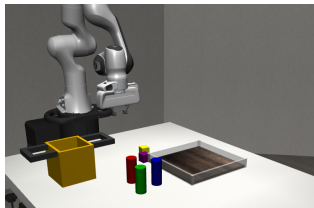
A Neuro-Symbolic Task and Motion Planning (TAMP) architecture bridging natural language reasoning with reactive physical execution.

Core Features

- ▶ **Ambiguity Resolution:** Proactively identify and resolve referential ambiguities in human prompts (e.g., "*Which cube?*") via multi-turn visual dialogue prior to physical execution.
- ▶ **Symbolic Verification:** Generate PDDL plans from semantic perception to guarantee causal correctness and eliminate syntactic and causal plan hallucinations given the grounded state.
- ▶ **Dynamic Behavior Tree:** Execute discrete symbolic plans using dynamically generated Behavior Trees, continuous condition monitoring, and auto-recovery fallbacks.
- ▶ **Hybrid Manipulation:** Combine Task-Space RRT for collision-free transit with RL (Proximal Policy Optimization) for compliant, contact-rich grasping.

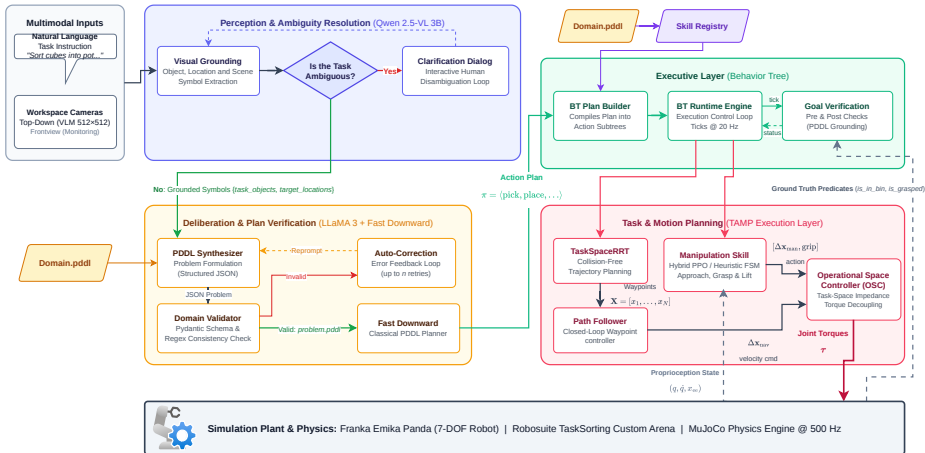
Scenario: *TaskSorting Arena*

The simulation environment is built on top of the Robosuite library with further customizations.



- ▶ **Robot Hardware:** 7-DoF Franka Emika Panda.
- ▶ **Scene Objects:** Cubes, cans and bins.
- ▶ **Physics Engine:** MuJoCo headless continuous simulation.
- ▶ **Perception:** Dual-view cameras (Top-Down + Frontal).

Neuro-Symbolic Planning architecture



Perception: Visual Grounding & Ambiguity Resolution

Grounds natural language human instructions in the observed scene and explicitly reasons about task ambiguity through multi-turn visual dialogue, resolving ungrounded assumptions before symbolic deliberation.

Ambiguity Resolution Pipeline

► VLM Visual Scene Grounding (Qwen 2.5-VL):

Parses the camera frame into entity schemas:

- task_objects: Manipulable items (e.g., cans, cubes).
- target_locations: Receptacles (e.g., sorting bin, pot).

► Ambiguity Detection: Evaluates query against detected scene entities:

- Flags referential ambiguity (e.g., multiple cubes present).
- Schema: AmbiguityReasoning(task_ambiguous, question).

► Human-in-the-Loop Clarification: Prompts the user with a targeted question, appends response to chat history, and disambiguates the scene.

► Symbols Grounding: Outputs clean symbols $\{\text{Objects}, \text{Locations}\} \subset O_{\text{obj}}, O_{\text{loc}}$ directly to the LLaMA-3 PDDL synthesizer.

Disambiguation Example

Human: "Put the cube in the container."

```
[Visual Scene Grounding]
task_objects: [yellow_cube, purple_cube]
target_locations: [sorting_bin, pot]
Ambiguity_Flagged: True
```

Robot: "Which cube should I move, the yellow or the purple one? And into the pot or the bin?"

Human: "The yellow cube into the pot."

```
[Symbols Grounding]
target: yellow_cube → (type obj)
destination: pot → (type location)
```

VLM Architectures: Dual-Encoder Foundations

Vision-Language Models (VLMs) integrate visual perception with linguistic reasoning by bridging continuous visual feature spaces and discrete language token spaces.

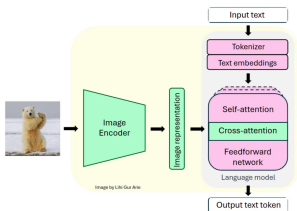


Figure: Standard Dual-Encoder VLM Architecture

Language Encoder (Transformer)

- ▶ Transforms discrete sequences into continuous numerical **embeddings**.
- ▶ Utilizes **self-attention mechanisms** to dynamically weigh token importance.
- ▶ **Decoders** auto-regressively generate the most statistically probable sequence.

Vision Encoder (CNN / ViT)

- ▶ Extracts hierarchical spatial features via Convolutional Neural Networks or Vision Transformers.
- ▶ ViTs segment images into non-overlapping patches, processing them as sequential tokens.

VLM Training Methodologies

Training fuses information from vision and language encoders, correlating visual features with text semantics to enable unified multimodal reasoning.

Contrastive Learning

Aligns modalities in a shared embedding space. The model minimizes the latent distance between matching image-text pairs while maximizing it for non-matching pairs.

Masked Modeling

Forces the model to reconstruct randomly obscured image patches or masked text tokens, learning robust contextual representations.

Generative Training

Optimizes the model to auto-regressively generate text conditioned on visual inputs (e.g., image captioning or visual question answering).

Pretrained Projection

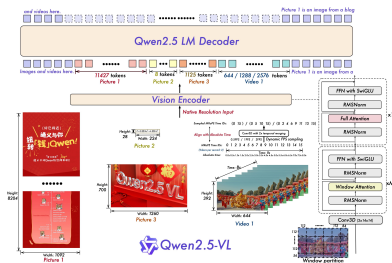
Freezes a pretrained LLM and Vision Encoder, training only a lightweight mapping network (e.g., MLP projector) to translate visual embeddings into the LLM's input space.

Qwen 2.5-VL Architecture

A state-of-the-art multimodal foundation model utilizing a dynamic-resolution Vision Transformer coupled with a robust language decoder.

Core Components

- ▶ **Qwen2.5 LM Decoder:** Advanced autoregressive language model for reasoning and generation.
- ▶ **Vision Transformer (ViT):** Uses 3D convolutions to efficiently process visual data into sequences of 14×14 patches.
- ▶ **Dynamic Preprocessor:** Adjusts image size dynamically to preserve native aspect ratios and extracts frames optimally for video inputs.



Design Rationale for this Architecture

- ▶ **Dynamic Resolution:** Preserves small object aspect ratios without distortion.
- ▶ **Local Real-Time Inference:** Runs locally via Ollama (1.25 s latency, zero API dependency).
- ▶ **High Spatial Fidelity:** Superior grounding of manipulable items and receptacle boundaries.

Symbolic Planning: PDDL Formulation

Symbolic Planning uses formal logic to find a sequence of actions that transition a system from an initial state to a target goal state. **PDDL** (Planning Domain Definition Language) is the standard modeling language for this, decoupling the robot's physical capabilities from specific task specifications to guarantee causal correctness before execution.

Formal Planning Tuple

$$\mathcal{P} = (\mathcal{D}, O, s_0, G)$$

► Planning Domain $\mathcal{D} = (\mathcal{L}, \mathcal{A})$ [Static]:

- $\mathcal{L}$: Predicates describing state relations.
- $\mathcal{A}$: Action schemas with preconditions and effects.
- *Fixed and verified offline* (robot capability bounds).

► Planning Problem Instance [Dynamic]:

- O : Grounded universe of objects and locations.
- s_0 : Completely grounded initial state from VLM.
- G : Target goal conditions from user task.
- *Synthesized online* via LLaMA 3 from perception.

Key Principle: Domain physics are fixed; only state and goal are queried online.

Manipulation Domain Definition (domain.pddl)

```
; Grounded State Predicates  $\mathcal{L}$ 
(holding ?obj)
(on-table ?obj)
(on ?obj ?loc)

; Robotic Action Schemas  $\mathcal{A}$ 
(:action pick
  :parameters (?o - obj)
  :precondition (and (on-table ?o)
                     (not (holding ?o)))
  :effect (and (holding ?o)
               (not (on-table ?o))))

(:action place
  :parameters (?o - obj ?loc - location)
  :precondition (holding ?o)
  :effect (and (on ?o ?loc)
               (not (holding ?o))))
```

Neuro-symbolic planning: PDDL Problem Synthesis

While domain physics $\mathcal{D}$ are fixed offline, the planning problem $\mathcal{P} = (\mathcal{D}, O, s_0, G)$ is formulated online. **LLaMA 3** translates visually grounded symbols and task intent into (O, s_0, G) to complete $\mathcal{P}$.

Problem Synthesis Workflow

- ▶ **1. Grounded Few-Shot Prompting:** Uses grounded entities ($O_{\text{obj}}, O_{\text{loc}}$), scene state s_0 , and task G alongside domain $\mathcal{D}$. Enforces strict typing ($O_{\text{obj}} \rightarrow \text{obj}$, $O_{\text{loc}} \rightarrow \text{location}$) and maps task goals to domain predicates.
- ▶ **2. Pydantic Validation, Auto-Repair & Compilation:** Validates LLM output against pydantic Problem schema, auto-correct undeclared symbols, and retries on predicate violations. `Problem.to_pddl()` deterministically serializes the verified instance into standard PDDL syntax.

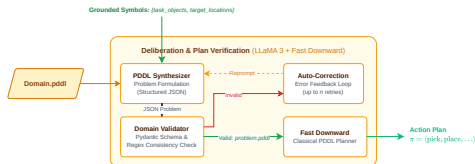
Key Guarantee: Structured JSON + Pydantic validation eliminates syntactic and schema hallucinations in the problem instance given the grounded state.

Generated Planning Problem (problem.pddl)

```
; Online problem instance  $\mathcal{P} = (\mathcal{D}, O, s_0, G)$ 
(define (problem manipulation-task)
  (:domain manipulation)
  (:objects
    red_can yellow_cube blue_can - obj
    sorting_bin pot - location
  )
  (:init
    (on-table red_can)
    (on-table yellow_cube)
    (on-table blue_can)
  )
  (:goal (and
    (on red_can sorting_bin)
    (on yellow_cube pot)
    (on blue_can sorting_bin)
  ))
)
```

Plan Verification: Fast Downward Solver

Fast Downward acts as the symbolic deliberation engine, using initial states s_0 , final goal G and Domain specification $\mathcal{D}$ to obtain a verified action sequences via **Heuristic Forward State-Space Search**.



Deliberation Guarantee: Computes verified causal action plans in < 0.08 s, eliminating syntactic and causal plan hallucinations given the grounded state.

Forward Search & Heuristics

- **Search Paradigm:** Progresses forward in state space $\mathcal{S}$:

$$s_0 \xrightarrow{a_1} s_1 \rightarrow \dots \rightarrow s_g \models G.$$

- **Direct Grounding:** Grounded s_0 from VLM makes forward state progression directly evaluable.
- **State Transitions:** Applies applicable action a ($\text{pre}(a) \subseteq s$):
 $s' = (s \setminus \text{del}(a)) \cup \text{add}(a)$.
- **Heuristic Engine (LAMA / h_{FF}):**

- **Lazy GBFS:** Greedy best-first search for high-speed plan discovery.
- **Dual Heuristics ($h_{FF} + h_{LM}$):** Delete-relaxation ($\text{del}(a) = \emptyset$) + causal Landmarks.
- **Preferred Operators:** Prunes search branching via helpful actions.

PDDL Problem Generation experiments

To bridge perceptual symbols and symbolic planning, the deliberation layer couples **Pydantic data models** with programmatic domain validation and a reflection loop to eliminate syntactic and schema hallucinations given the grounded state.

Structured Pipeline & Pydantic Schemas

- ▶ **Synthesizer Class** (`AmbresStructured`):
`generate_problem_json(...)` queries LLaMA 3 with $(s_0, G, O_{\text{obj}}, O_{\text{loc}}, \mathcal{D})$ under few-shot JSON constraints.
- ▶ **Data Models** (`src.llm2pddl.problem`):
 - ▶ **Problem**: Declares objects, init, and goal.
 - ▶ **PDDLObject**: Type enforcement (`obj, location`).
 - ▶ **Predicate**: Structured facts with arity validation.
- ▶ **Compilation** (`Problem.to_pddl()`):
Deterministically convert the JSON payload schema into standard PDDL formatted text.

Closed-Loop Validation & Auto-Repair

- ▶ **Validation Function**:
`generate_problem_with_retry(..., \mathcal{D})` verifies predicates and types against $\mathcal{D}$ via regex extraction.
- ▶ **Handled Failure Modes**:
 - ▶ *Predicate Hallucination*: e.g., $(\text{in } ?x \text{ ?bin}) \notin \mathcal{D}$.
 - ▶ *Undeclared Symbols*: Missing object in objects.
 - ▶ *Type Inversion*: Receptacle labeled as obj.
- ▶ **Auto-Healing & Reflection**:
 - ▶ Inferred auto-declaration for missing receptacles.
 - ▶ Re-prompts LLaMA 3 with exact error traces (max 5 retries).

PDDL Problem Generation experiments

Empirical validation across **28 simulation trials** demonstrates that closed-loop PDDL synthesis achieves 96.4% feasibility and reliably shields the executive layer.

Empirical Benchmark

Evaluation Metric	Result
Total Logged Trials	28
1st-Pass Solver Feasibility	19 / 28 (67.9%)
Success after Feedback Loop	27 / 28 (96.4%)
Fast Downward Parser Errors	0% (0 / 28)
Avg. Symbolic Solve Time	0.103 s
Max Sequential Objects	4 (8 plan steps)

- ▶ **67.9% First-Pass Feasibility:** Raw LLaMA 3 output before verification.
- ▶ **96.4% Closed-Loop Success:** Feedback reflection repairs 8/9 initial errors within 1–2 retries.

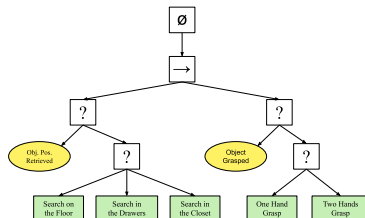
Impact on the Executive Layer

- ▶ **Causal Feasibility Verification:**
Fast Downward verifies task feasibility *prior* to actuation; unreachable goals fail safely offline.
- ▶ **Plan-to-BT Compiler:**
`build_bt_from_pddl_plan(plan, adapter)` compiles verified plans $\pi = \langle a_1, \dots, a_N \rangle$ into reactive Behavior Trees.
- ▶ **Executive Shielding:**
The Behavior Tree is shielded from malformed symbols, broken preconditions, or causally invalid action sequences given the grounded state.

Bridge to Execution: Formal symbolic deliberation ensures causal and syntactic plan consistency given the grounded state before handoff to Behavior Tree execution.

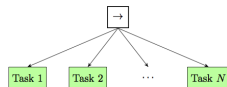
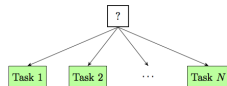
Executive Layer: Behavior Trees

- ▶ A **Behavior Tree** is a directed tree with nodes classified as root, control flow, or execution (tasks).
- ▶ Execution starts from the root, which sends *ticks* to its children at a set frequency.
- ▶ A node returns one of three statuses: **Running**, **Success**, or **Failure**.



Control Flow Nodes

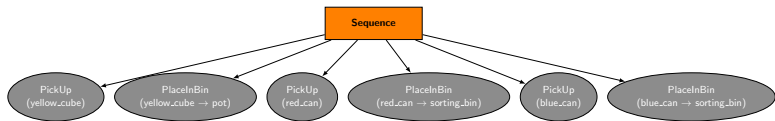
- ▶ **Selector:** A fallback node returns **Success** if at least one child do not fail. If every child fail return **Failure**. Otherwise **Running**.
- ▶ **Sequence:** Executes children in order, returning success if everyone succeeds. Returns **Failure** or **Running** immediately if a child fails or is still running.



How to build Behavior Trees: `py_trees` library

`py_trees` is a Python library used to construct Behavior Trees efficiently. It relies primarily on two base classes: **Behaviours** and **Composites**.

- ▶ **Behaviour (Leaf):** The smallest element in a tree. Behaviours typically represent a **condition** (e.g., *Am I hungry?*) or an **action** (e.g., *Buy chocolate cookies*).
- ▶ **Composites (Control Flow):** Responsible for directing the path traced through the tree during a tick.
 - ▶ **Selectors:** Fallback decision-makers.
 - ▶ **Sequences:** Order-dependent factories.
 - ▶ **Parallels:** Concurrent execution factories.



BT Builder: Parsing and Sequence Compilation

1. Parsing PDDL Plan

(`parse_pddl_action`) Cleans raw planner text into structured Python data:

- ▶ **Filters Clutter:** Discards comments starting with ; (e.g., ; cost = 2) and blank lines.
- ▶ **Strips Planner Syntax:** Removes step prefixes (e.g., 0:) and outer parentheses (...).
- ▶ **Extracts Tokens:** Normalizes action name to lowercase and collects arguments:

```
"0: (place yellow_cube pot)" → ("place",  
    ["yellow_cube", "pot"])
```

2. Plan to Sequence

(`pddl_plan_to_sequence`) Converts the parsed action list into an executable Behavior Tree Sequence:

- ▶ **Skill Dispatch:** Uses `SKILL_REGISTRY` to map actions to robot skills ("pick", "place").
- ▶ **Argument Binding:** Passes simulator environment and object targets (object, asset).

Compilation Pipeline Example

1. Planner Output (Fast Downward):

```
; plan generated by fast downward  
0: (pick yellow_cube)  
1: (place yellow_cube pot)  
; cost = 2
```

⇓ *parsed & compiled into BT*

2. Executable PyTrees Sequence:

```
[~] pddl_plan_sequence (memory=True)  
|-- PickUp(object='yellow_cube')  
'-- PlaceInBin(object='yellow_cube',  
    asset='pot')
```

Design Rationale

Cleanly decouples **symbolic task planning** (Fast Downward) from the **physical robot control** (Motion planning).

BT Builder: Goal Checking and Tree Generation

3. Goal-Guarded Wrapper

(wrap_with_goal_check) A plain sequence is open-loop. If an object drops, it fails blindly. We wrap it in a reactive fallback loop:

- ▶ **Pre-Check (GoalCheck_Pre)**: Verifies if goal is already met before moving. If yes → succeeds immediately (no redundant actions).
- ▶ **Selector (memory=False)**: Re-evaluates continuously on every tick for dynamic preemption.
- ▶ **Post-Check (GoalCheck_Post)**: Confirms physical completion after the sequence finishes.
- ▶ **Retry Loop (Retry)**: If the grasp slips and post-check fails, automatically retries the plan up to N times.

4. Main Entrypoint

(build_bt_from_pddl_plan) Single builder API providing two execution modes:

- ▶ **wrap_goal_check=False**: Raw Sequence (open-loop baseline).
- ▶ **wrap_goal_check=True**: Self-healing tree with retry loop.

Goal-Guarded Tree Architecture

```
[?] goal_guarded_root (Selector)
|-- [>] GoalCheck_Pre (already met?)
'-- [R] retry_until_goal (max=3)
    '-- [->] attempt (Sequence)
        |-- [->] pddl_plan_sequence
        |   |-- PickUp(yellow_cube)
        |   '-- PlaceInBin(yellow_cube->pot)
        '-- [>] GoalCheck_Post (verified?)
```

Failure Recovery Example (Cube Drop)

1. Robot picks up cube, but gripper slips mid-air.
2. Sequence finishes → GoalCheck_Post checks pot.
3. Cube is missing → returns FAILURE.
4. Retry catches failure → re-runs the plan → task completed successfully!

BT Nodes: Skills & Conditions

The executive layer instantiates action skills and condition monitors as modular `py_trees` nodes, bridging discrete PDDL plans to continuous 20 Hz robot control.

Action Skills (`skills.py`)

- ▶ **Base Architecture (Skill):**
Subclasses `Behaviour`. Non-blocking 20 Hz tick: returns `RUNNING`, `SUCCESS` on `is_done()`, or `FAILURE` on timeout. Emits 7-DOF OSC commands $a \in \mathbb{R}^7$.
- ▶ **Pick Primitive (MotionPlanningPickUp):**
4-stage FSM: `follow_path` (RRT to hover) $\rightarrow$ `descend` (linear approach) $\rightarrow$ `grasp` (finger closure) $\rightarrow$ `lift` (clearance).
▶ *Hybrid PPO Variant:* Replaces descent/grasp with learned RL policy.
- ▶ **Place Primitive (MotionPlanningPlaceInBin):**
2-stage FSM: `follow_path` (RRT transit with locked grip) $\rightarrow$ `release` (open gripper over container).

Condition Nodes (`conditions.py`)

- ▶ **Post Conditions Checking(conditions.py):**
Evaluates actions `add(s)` effects at zero latency:
 - `on(?x, ?y)  $\leftrightarrow$  env.is_in_bin(obj, target)`
 - `holding(?x)  $\leftrightarrow$  env.is_grasped(obj)`
 - `on-table(?x)  $\leftrightarrow$   $\neg$ grasped  $\wedge$   $\neg$ in_bin`
- ▶ **Goal Monitor (PDDLGoalCheck):**
Parses conjunctive goals $\bigwedge g_i$ from `problem.pddl`. Evaluates pre/post conditions to skip met goals or trigger retry.

Execution & Runtime Tree

The runtime engine executes the compiled Behavior Tree via a 20 Hz tick loop, continuously verifying action post-conditions, dispatching motion skills, and recovering from disturbances.

Runtime Execution Pipeline

- ▶ **20 Hz Tick Loop (py-trees):**
Synchronous active-leaf evaluation; steps MuJoCo with 7-DOF OSC commands $a \in \mathbb{R}^7$.
- ▶ **Closed-Loop Reactive Flow:**
 - *Pre-Check:* GoalCheck.Pre skips met goals (0 redundant steps).
 - *Skill Dispatch:* Steps TaskSpaceRRT waypoints or PPO policy.
 - *Post-Check:* Holding / On verify physical effect immediately.
 - *Recovery:* Action failure triggers retry_until_goal loop.
 - *Post-Check:* GoalCheck.Post verify if all the goal predicates are true.
- ▶ **Real-Time Telemetry:** Dual-camera HUD overlays active leaf, skill status, and ticks.

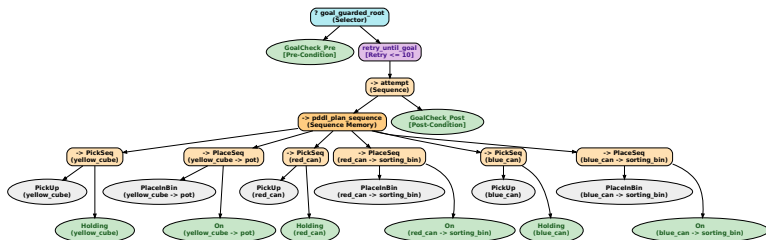
Goal-Guarded Tree Architecture

```
[?] goal_guarded_root (Selector)
|-- [>] GoalCheck_Pre (already met?)
'-- [R] retry_until_goal (max=10)
    '-- [->] attempt (Sequence)
        |-- [->] pddl_plan_sequence (Memory)
        |   |-- [->] PickSeq(yellow_cube)
        |   |   |-- PickUp(yellow_cube)
        |   |   |-- [>] Holding(yellow_cube)
        |   |   |-- [->] PlaceSeq(cube->pot)
        |   |   |-- PlaceInBin(cube->pot)
        |   |   |-- [->] On(cube->pot)
        |   '-- [->] GoalCheck_Post (verified?)
```

- ▶ **Selector [?]:** Skips plan if met; falls back to retry loop.
- ▶ **Sequence [->]:** Action $\rightarrow$ Post-check; halts on slip/drop.
- ▶ **Retry [R]:** Autonomous recovery loop (≤ 10 attempts).

Reactive Robustness: Closed-loop verification of positive post-conditions halts flawed execution immediately, enabling autonomous recovery without cascading errors.

Main Scenario: Generated Behavior Tree



1. Goal Conditions Check

- **GoalCheck_Pre**: skips action if goal met (0 latency).
- **GoalCheck_Post**: verifies overall task formula $\mathcal{G} = \bigwedge_i on(x_i, loc_i)$.

2. Action Sequence

- **Pick**: pick action and effects conditions $Holding(x)$ ($\Delta z > 2.5$ cm) checking.
- **Place**: place action and effects conditions $On(x \rightarrow loc)$ (containment) Checking.
- Halts execution immediately on FAILURE conditions (slip/miss).

3. Reactive Recovery

- **retry_until_goal** ≤ 10 attempts.
- Intercepts disturbance $\rightarrow$ re-attempts plan rather than blindly continuing.

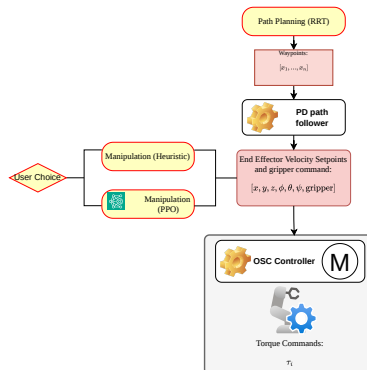
Reactive Layer: Motion and Path Planning

Reactive Layer Roles

- ▶ **Symbolic-to-Continuous Grounding:** Translates discrete BT skills (pick, place) into continuous robot motion.
- ▶ **Perception:** Accounts for physical geometry, obstacles, and contact dynamics ignored by PDDL.
- ▶ **Motor Control:** Operates at 20 Hz with real-time state feedback from MuJoCo Simulator Environment.

Hierarchical Motion Decomposition

- ▶ **1. Path Planning (Task Space RRT):** Solves pure geometric obstacle avoidance in free space ($\mathbb{R}^3$).
- ▶ **2. Trajectory Tracking:** Closed-loop proportional tracking of Cartesian waypoints $[x_1, \dots, x_n]$ generated by the RRT.
- ▶ **3. Manipulation (PPO / Heuristic):** Governs fine contact dynamics, compliant grasping, and lifting.
- ▶ **4. Low Level Controller (OSC):** Takes as input the cartesian velocities from the Tracking and Manipulation modules and output the joint torques τ_i needed to control the robot.



Path Planning: RRT

Rapidly Random Trees (RRT) is a Probabilistic Path Planning method used to efficiently construct a path in continuous spaces.

Algorithm Goal Oriented RRT Algorithm

```
1:  $G.\text{init}(x_{\text{start}})$ 
2: for  $i = 1 \dots K$  do
3:    $p = \text{UniformRandom}(0, 1)$ 
4:   if  $p \leq p_{\text{goal}}$  then
5:      $x_{\text{rand}} \leftarrow x_{\text{goal}}$ 
6:   else
7:      $x_{\text{rand}} \leftarrow \text{RandPose}()$ 
8:   end if
9:    $x_{\text{nearest}} \leftarrow \text{NearestVertex}(x_{\text{rand}}, G)$ 
10:   $x_{\text{new}} \leftarrow \text{NewConf}(x_{\text{nearest}}, x_{\text{rand}}, \Delta x)$ 
11:  if  $\text{CollisionFree}(x_{\text{nearest}}, x_{\text{new}})$  then
12:     $G.\text{add\_vertex}(x_{\text{new}})$ 
13:     $G.\text{add\_edge}(x_{\text{nearest}}, x_{\text{new}})$ 
14:    if  $x_{\text{new}} \in X_{\text{goal}}$  then return Path Found
15:  end if
16: end if
17: end for
18: return  $G$ 
```

The goal oriented version samples the goal point with a given probability p while a random point with probability $1 - p$.

Trajectory Tracking

- ▶ **Cartesian Setpoints:** The waypoints generated by the task-space RRT are Cartesian poses used as setpoints for a **Proportional (P) Controller**.
- ▶ **Tracking Error:** At each time step, the position error is defined as:

$$\mathbf{e} = \mathbf{p}_{\text{target}} - \mathbf{p}_{\text{EEF}}$$

- ▶ **Control Law:** The controller outputs velocity commands bounded by a maximum speed:

$$\mathbf{v}_{\text{cmd}} = \text{clip}(K_p \mathbf{e}, -v_{\text{max}}, v_{\text{max}})$$

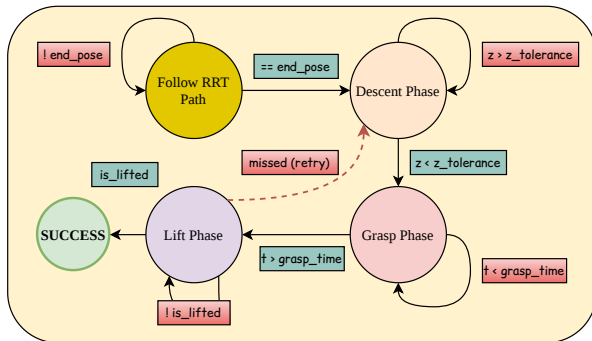
which are sent to the underlying operational space controller.

- ▶ **Application:** Used specifically during the **approach phase** in the pick action and during the **place action**. The place skill itself is completely based on the RRT planner and the trajectory tracker, while the pick skill is composed by more modules and need further explanation.

Manipulation: Heuristic Approach

Consists of a **Staged FSM**, that executes the following phases:

1. **RRT Planning:** Traverse collision-free waypoints to hover pose ($== \text{end_pose}$).
2. **Descent Phase:** Descend vertically until reaching grasp height ($z \leq z_{\text{tol}}$).
3. **Grasp Phase:** Actuate gripper firmly for set duration ($t > t_{\text{grasp}}$).
4. **Lift Phase:** Elevate object to safe height to verify pick. If it fails, it will retry from the Descent phase.



Manipulation: Proximal Policy Optimization

Proximal Policy Optimization is an on-policy actor-critic Reinforcement Learning method. PPO updates policy parameters θ_{k+1} by maximizing the **Clipped Surrogate Objective**:

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

where the probability ratio is defined as:

$$r_t(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_k}(a_t | s_t)}$$

- ▶ **Clipped Objective:** The hyperparameter $\epsilon \approx 0.2$ penalizes moving $r_t(\theta)$ outside $[1 - \epsilon, 1 + \epsilon]$, preventing destructively large policy updates.
- ▶ **Actor-Critic Architecture:** Jointly optimizes a policy network $\pi_{\theta}(a|s)$ and a state-value baseline $V_{\phi}(s)$ using Generalized Advantage Estimation (GAE).

Manipulation: Observation Space

TaskSpaceRRT provides coarse global navigation to a pre-grasp hover pose (~ 18 cm above the object). The **PPO policy** then takes over using a minimal, translation-invariant **11D observation vector**:

$$\mathbf{o}_t = \left[\underbrace{\Delta \mathbf{p}}_{\mathbb{R}^3}, \underbrace{\mathbf{q}_{\text{grip}}}_{\mathbb{R}^2}, \underbrace{\dot{\mathbf{q}}_{\text{grip}}}_{\mathbb{R}^2}, \underbrace{z_{\text{rel}}^{\text{EEF}}}_{\mathbb{R}^1}, \underbrace{\Delta z_{\text{obj}}}_{\mathbb{R}^1}, \underbrace{c_{\text{grasp}}}_{\{0,1\}}, \underbrace{m_{\text{geom}}}_{\{0,1\}} \right]^T \in \mathbb{R}^{11}$$

Kinematics & Spatial Relations

- ▶ $\Delta \mathbf{p} = \mathbf{p}_{\text{obj}} - \mathbf{p}_{\text{EEF}}$: 3D relative position vector (translation invariance across table).
- ▶ $z_{\text{rel}}^{\text{EEF}} = p_{\text{EEF},z} - z_{\text{table}}$: Table clearance to avoid surface collisions.
- ▶ $\Delta z_{\text{obj}} = p_{\text{obj},z} - p_{\text{obj},z}^{(0)}$: Vertical displacement measuring lift progress.

Gripper and Auxiliary States

- ▶ $\mathbf{q}_{\text{grip}}, \dot{\mathbf{q}}_{\text{grip}}$: Left/right finger joint positions and velocities (contact stall detection).
- ▶ $c_{\text{grasp}} \in \{0, 1\}$: Bilateral force-closure flag (Robosuite contact normal check).
- ▶ $m_{\text{geom}} \in \{0, 1\}$: Shape token (0: cube, 1: can) enabling multi-geometry conditioning.

Manipulation: Reward Function

The agent is trained with a multi-stage continuous-discrete reward guiding the transition from approach $\rightarrow$ grasp $\rightarrow$ lift:

$$R_t = r_{\text{reach}} + r_{\text{grip}} + r_{\text{grasp}} + r_{\text{lift}} - c_{\text{step}} + r_{\text{terminal}}$$

Continuous Shaping Terms

- **Smooth Reaching** ($d = \|\Delta \mathbf{p}\|_2$):
 $r_{\text{reach}} = 1.0 - \tanh(10.0 \cdot d)$
- **Grasp Maintenance:**
 $r_{\text{grasp}} = 2.0 \cdot \mathbb{I}(\text{grasped})$
- **Conditioned Vertical Lift:**
 $r_{\text{lift}} = 20.0 \cdot \max(0, \Delta z_{\text{obj}}) \cdot \mathbb{I}(\text{grasped})$
(Avoids lifting with an empty gripper)
- **Alive Cost:** $c_{\text{step}} = 0.01$ (incentivizes speed)

Action Shaping & Terminals

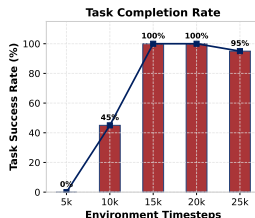
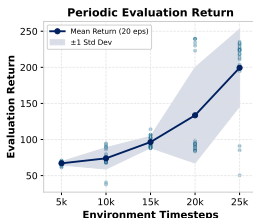
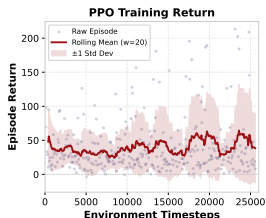
- **Gripper Timing Incentive** r_{grip} :

$$r_{\text{grip}} = \begin{cases} +0.5 & \text{if } d < 4 \text{ cm} \wedge a_g > 0 \\ -0.25 & \text{if } d < 4 \text{ cm} \wedge a_g \leq 0 \\ -0.2 & \text{if } d \geq 4 \text{ cm} \wedge a_g > 0.2 \\ 0 & \text{otherwise} \end{cases}$$

Prevents premature closure before alignment.

- **Terminal Sparse Rewards:**
 $+15.0$ if $\Delta z_{\text{obj}} \geq 6 \text{ cm} \wedge \text{grasped}$ (**Success**)
 -2.0 if $p_{\text{obj},z} < z_{\text{table}} - 4 \text{ cm}$ (**Drop**)

Manipulation: PPO Training & Evaluation



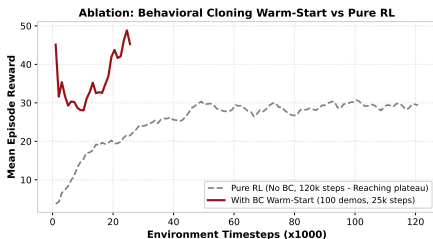
Training Return

- **Stochastic Action Noise:** Persistent Gaussian exploration noise (with standard deviation $\sigma \approx 1.0$) perturbs micro-contacts during training (~ 50 avg).
- **Peak Episodes (≈ 200 – 230):** When random noise aligns perfectly, trajectories achieve full force closure and lift with terminal bonus (+15).

Evaluation and Task Completion

- **Deterministic Evaluation:** Setting $\sigma = 0$ and evaluating via the distribution mean μ effectively removes action jitter, revealing the true learned policy return of **199.4** (near theoretical max ~ 200).
- **Success Rate Ascent:** Climbs rapidly from **0% $\rightarrow$ 45% $\rightarrow$ 95%** robust task completion.

Manipulation: Behavioral Cloning Warm-Start



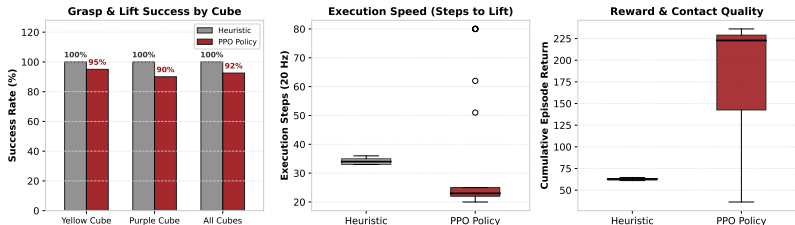
Synthetic Demonstrations

- ▶ Generated via a closed-loop 3-stage state-machine using simulation ground-truth poses.
- ▶ 100 demos ($\sim 3,000$ transitions) pre-trained for 150 epochs via supervised MSE loss.

Pure RL vs BC-RL

- ▶ **Pure RL (120k steps):** Gets stuck in the "*reaching local minimum*" (≈ 29 reward), never discovering the precise descent-grasp-lift sequence.
- ▶ **BC Warm-Start (25k steps):** Initializes policy in the basin of attraction, achieving immediate task competence in $5\times$ fewer steps.

Manipulation: Heuristic vs PPO Comparison



Manipulation Performance

- **Generalization:** PPO achieves **95%** on **Yellow Cube** (consistent with training) and **90%** on **Purple Cube** (zero-shot transfer), averaging **92.5%**.
- **Micro-Manipulation Latency:** PPO median is **1.10 s** (22 steps) vs. Heuristic **1.72 s** (34.3 steps) (PPO mean is 33.5 due to rare retry outliers).
- **Eliminates Dwell Timers:** Heuristic relies on fixed post-grasp dwell timers; PPO senses force closure and lifts dynamically upon contact.

Trade-Offs

- **Dynamic Impedance:** PPO modulates continuous velocities based on real-time force closure rather than rigid fixed gains ($K_p = 10$).
- **Privileged vs Reactive:** Heuristic requires exact object coordinates; PPO relies only on local $\Delta \mathbf{p}$ and joint kinematics.
- **Safety Guard:** Hybrid pick node falls back to heuristic if step budget (≤ 90) expires.

Experimental Setup: Simulation Environment



Robosuite TaskSorting Arena

- ▶ **Robot:** 7-DoF Franka Emika Panda with parallel-jaw gripper (20 Hz OSC control).
- ▶ **Physics:** MuJoCo headless physics engine with compliant contact dynamics.
- ▶ **Dual Viewpoints:** Orthographic *top-down* (VLM grounding) + perspective *frontview* (monitoring).
- ▶ **Objects:** 5 pickable items (yellow/purple cubes, red/blue/green cans).
- ▶ **Containers:** Sorting bin (30×30 cm) + handled pot (collision obstacles).

End-to-End TAMP Pipeline Execution

Task Mission: *"Put the yellow cube in the pot, then put the red can in the sorting bin, then put the blue can in the sorting bin."*

1. Perception & Plan

- ▶ **Qwen 2.5-VL:** Scene Detection.
- ▶ **Llama 3:** PDDL synthesis.
- ▶ **Fast Downward:** < 0.1 s plan.

2. BT Compilation

- ▶ Compiled py_trees reactive tree.
- ▶ Pre-conditions & fallback nodes.
- ▶ 20 Hz continuous monitoring.

3. Motion & Control

- ▶ **TaskSpaceRRT:** Paths (~ 30 ms).
- ▶ **Hybrid PPO:** Grasp/lift (≈ 1.7 s).
- ▶ **Placement:** Waypoint tracking.



▶ Heuristic Trial (simulation_trial_heuristic.gif)



▶ PPO Policy Trial (simulation_trial_ppo.gif)

▶ GitHub Repo: [vincip99/.../tree/master](https://github.com/vincip99/tamp/tree/master)

Experimental Results: Main Scenario Performance

Pipeline Layer	Module / Algorithm	Latency / Steps
Visual Grounding	Qwen 2.5-VL (3B)	1.25 s
PDDL Formulation	LLaMA 3 (8B)	0.65 s
Symbolic Planning	Fast Downward (A^*)	0.08 s (6 causal actions)
Motion Planning	TaskSpaceRRT	0.03 s / transit leg (6 legs)
Manipulation (Pick)	Macro Transit + PPO Grasp*	79–132 steps / pick
Reactive Executive	Behavior Tree (py_trees)	445 ticks (22.2 s)
End-to-End System	Complete Architecture	24.2 s Total (100% Success)

Planning and Execution

- **Causal Plan:** $\langle \text{pick}(\text{cube}) \rightarrow \text{pot}, \text{pick}(\text{red}) \rightarrow \text{bin}, \text{pick}(\text{blue}) \rightarrow \text{bin} \rangle$.
- **Pick Latency (*)**: End-to-end pick combines macro RRT approach transit (~ 55 – 100 steps) and micro PPO grasp (20 – 25 steps, Slide 29).
- **Total Duration**: Completed in 445 ticks (22.2 s) without timeouts.

Robustness

- **Obstacle Clearance**: 0 collisions with bin rims or pot handles across all 6 transit trajectories.
- **Grasp Stability**: Zero object slip or drop events during transit; firm force closure maintained throughout.

Experimental Results: 28-Task Benchmark Performance

Deliberation and execution benchmark across 28 manipulation tasks:

Benchmark Complexity Tier	Trials	Plan Length	1st-Pass Feas.	Post-Refl. Feas.	Task Execution Success
Tier 1: Single-Object Relocation	5	2 steps	80.0% (4/5)	100.0% (5/5)	100.0% (5/5)
Tier 2: Two-Object Sequential	6	4 steps	50.0% (3/6)	100.0% (6/6)	100.0% (6/6)
Tier 3: Three-Object Chained	7	6 steps	57.1% (4/7)	85.7% (6/7)	85.7% (6/7) [†]
Tier 4: Four-Object High-Complexity	3	8 steps	66.7% (2/3)	100.0% (3/3)	100.0% (3/3)
Tier 5: Ambiguity & Lexical Synonyms	7	2–6 steps	85.7% (6/7)	100.0% (7/7)	100.0% (7/7)
Total Benchmark Dataset	28	Up to 8 steps	67.9% (19/28)	96.4% (27/28)	96.4% (27/28)

[†] End-to-end task completion rate. The single Tier-3 failure occurred at PDDL generation (Trial 15); physical execution succeeded on 100% of dispatched plans (6/6 in Tier 3, 27/27 overall).

LLM Failure Modes & Auto-Repair

- **Schema Validation (21.4%):** Syntax / phrasing errors caught by Pydantic; repaired on Retries 1–2.
- **State Incompleteness (Trial 9):** Initial predicate omission repaired via compiler trace feedback.
- **Duplicate Declaration (Trial 15):** LLaMA 3 repeated receptacle declaration in (:objects), caught by solver.
- **Auto-Repair Gain:** Lifted feasibility from 67.9% → 96.4% (8/9 initial errors resolved).

Manipulation: PPO Failures & BT Fallback

- **PPO Competence: 92.5%** pick success on cubes (95% Yellow trained, 90% Purple zero-shot).
- **PPO Failure Modes:** Out-of-distribution tall cylinder geometry and micro-contact roll cause step budget timeout (≥ 90 steps).
- **Fallback Guarantee:** Behavior Tree fallback triggers deterministic FSM upon timeout → 100% physical execution success on all dispatched plans (6/6 in Tier 3, 27/27 overall).

Conclusions & Future Perspectives

A hierarchical 3-layer architecture bridging multimodal foundation models with formal symbolic deliberation and closed-loop reactive manipulation.

Core Contributions

- ▶ **Neuro-Symbolic Deliberation:** Autonomous extraction of (s_0, G) from RGB frames and language, fortified by interactive disambiguation and auto-repair (96.4% solver feasibility).
- ▶ **Formal Fast Downward Gatekeeper:** Guarantees causal plan validity in < 0.08 s before physical dispatch.
- ▶ **Reactive Executive Layer:** Behavior Trees compile action sequences into tick-based subtrees with continuous pre/post-condition monitoring.
- ▶ **Robust Hybrid Control:** TaskSpaceRRT ensures collision-free transit, while BC-bootstrapped PPO achieves reliable contact-rich grasping.

Future Developments

- ▶ **Domain & Problem Co-Synthesis:** Moving beyond fixed domain templates toward online predicate invention and automated action schema discovery.
- ▶ **Skill Registry Augmentation:** Expanding the primitive library to articulated objects and different actions.
- ▶ **Unified RL/IL for TAMP:** End-to-end hierarchical policy learning bridging task logic with real-time visuomotor control.
- ▶ **Sim-to-Real Transfer:** Hardware validation on physical 7-DOF Franka Emika Panda setups.

Key References



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